

SYSTEM REQUIREMENTS SPECIFICATION

UPVC FACTORY ORDER AUTOMATION & TRACKING SYSTEM

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1. EXECUTIVE OVERVIEW & SCOPE

This System Requirements Specification (SRS) formalizes the operational requirements extracted from factory notes for developing a custom **Order Automation, QR Scanning, and Workflow Execution System** for uPVC window manufacturing operations. The primary objective is to streamline order tracking, automate document management, eliminate manual pricing delays, and strictly enforce dispatch payment verification via integrated WhatsApp notifications and QR scanning terminals.

Core Operational Mandate: Link shop-floor QR scanning (Assembly & Dispatch) with centralized order status tracking, automated document ingestion (Quotes, Confirmations, Cutting Lists), glass procurement scheduling, and automated client payment alerts prior to delivery.

2. SYSTEM MODULES & HANDWRITTEN NOTES CROSS-REFERENCE

NOTE #	SYSTEM MODULE	SOURCE REQUIREMENT TEXT	IT FUNCTIONAL DELIVERABLE
Note 1	Assembly QR Scan	QR sticker to be placed on windows during assembly and scanned so that we know which frame/shutter has finished production.	Shop-floor scanner UI to update individual frame status to FABRICATION_COMPLETED .
Note 2	Dispatch QR Scan	Same QR will be scanned during dispatch to know what is being dispatched.	Dispatch gate terminal scanning to match items against manifest and verify release eligibility.
Note 3	Central Tracking Engine	Software integration with the scanning system to track status of order.	Real-time event sync engine providing continuous order status visibility across factory stages.
Note 4	Document Repository	Option to upload confirmation, quotation, final cutting list for tracking.	Multi-file upload module attached to Order ID (Confirmations, Quotes, EvA ERP Cutting Lists).
Note 5	Order ID & Location Scheme	Maintain an order id (e.g., C - Chavan Bhau, R - Rayal) followed by numbers & location codes (e.g., R253 PN = Rayal #253, Pune).	Standardized Order ID Generator logic: [Client_Prefix][Seq_No][Location_Code] .
Note 6	Glass Schedule Tracker	Glass dates are uncertain so there should also be option to track glass dates/status.	Independent Glass Procurement Tracker handling vendor lead times and fitting dates.
Note 7	Quote Ingestion & Approval	Feature to upload finished quote with window qty, sqft, glass type & client details. Notify Uday Sir; unpriced quotes take priority.	Quote parser + Rate approval workflow with priority queue & automated alert to Uday Sir.
Note 8	WhatsApp Payment Alerts	Utilizing the QR scanner, clients should be notified via whatsapp for balance payments.	Automated WhatsApp API trigger on QR scan to send payment link if balance is pending.

3. DETAILED FUNCTIONAL SPECIFICATIONS

MODULE 1: Standardized Order Identification & Location Coding	[FR-ORD-01]
Business Logic (Note 5): The system must generate and enforce a standardized Order Identifier incorporating client account prefixes, sequential order numbering, and project site location codes. Syntax Format: [CLIENT_PREFIX][SEQUENCE_NO][LOCATION_CODE] Client Prefixes: C = Chavan Bhau Orders R = Rayal Orders Configurable lookup table for new clients. Sequence Number: Sequential integer assigned upon order creation (e.g., 253). Location Suffix: 2-3 letter site geographic code (e.g., PN = Pune, MUM = Mumbai, WAD = Wadala). Example Canonical Output: R253 PN (Rayal Order #253 assigned to Pune location).	

Document Management (Note 4 & Note 7): Attach and store files against each Order ID:

1. **Quotation Document:** Initial estimated pricing document.
2. **Finished Confirmation PDF:** Extracted fields include Window Qty, Total SQFT, Glass Type, Profile Type, and Client Details.
3. **Final Cutting List:** Profile cutting and sizing sheet exported from EvA ERP software.

Unpriced Quotation Priority & Notification Trigger (Note 7):

- When a quotation is uploaded where rates/pricing have not been populated, the system automatically flags the order as **HIGH PRIORITY - RATE AWAITED**.
- Unpriced quotes must automatically float to the top of the management dashboard.
- An automated push notification / message must be sent directly to **Uday Sir** for rate approval before quotation release.

Frame-Level QR Generation & Assembly Tracking (Note 1 & Note 3):

- Upon cutting list approval, the system generates unique printable 2D QR stickers for every frame and sash (e.g., **R253-PN-W01-F**).
- QR stickers are physically attached to frames during assembly.
- **Assembly Station Scanning:** Factory workers scan the QR code using a handheld barcode terminal upon completing assembly.
- **System Action:** Updates frame status in real-time to **FABRICATION_COMPLETED** with timestamp and operator ID.

Supplier Date Uncertainty Management (Note 6): Due to lead-time variations from external glass toughening/DGU suppliers, the software provides a dedicated Glass Tracking tab.

Tracked Milestone Fields:

- **GLASS_ORDERED** Date requisition sent to glass processor.
- **EXPECTED_DELIVERY** Estimated dispatch date provided by vendor.
- **GLASS_RECEIVED** Actual arrival date at window plant.
- **GLASS_FITTED** Date glass glazing is installed into window frame.

System displays visual delay warnings on the main dashboard if current date exceeds **EXPECTED_DELIVERY** without receipt confirmation.

Dispatch Gate Verification (Note 2 & Note 3): Loading dock operators scan each window frame's QR code prior to truck loading. The system verifies scanned frames against the order manifest.

Automated WhatsApp Balance Payment Alert (Note 8):

- Upon scanning frames during dispatch check, the system evaluates the financial balance ledger for the Order ID.
- **Trigger Logic:** If a balance payment is outstanding (e.g. 30% pre-dispatch payment), the scan instantly triggers an automated **WhatsApp API Message** to the client's registered mobile number.

• **WhatsApp Message Template:**

"Dear [Client Name], your uPVC window order [Order ID: R253 PN] is being loaded for dispatch!
Outstanding balance: ₹[Amount]. Please process payment to complete dispatch clearance. Payment Link:
[UPI/Bank Link]"

- **Gate Pass Hard Lock:** The system locks Gate Pass printing until Accounts confirms payment clearance.

4. TECHNICAL DATA SCHEMA & INTEGRATION ARCHITECTURE

4.1 Relational Database Table Schema

TABLE NAME	PRIMARY ATTRIBUTES / COLUMNS	RELATIONSHIPS & INTEGRITY RULES
orders	order_id (VARCHAR, PK, e.g. 'R253 PN'), client_name, location_code, total_window_qty, total_sqft, glass_type, profile_type, total_amount, balance_due, current_status	Master table. PK formatted as [Prefix][Seq][Loc].
order_documents	doc_id, order_id (FK), doc_type ('QUOTATION', 'CONFIRMATION', 'CUTTING_LIST'), file_path, uploaded_at, rate_approved (BOOLEAN)	Links PDF/Excel files to Order ID. Stores rate approval status.
window_frames	frame_qr_code (PK, e.g. 'R253-PN-W01-F'), order_id (FK), frame_label, assembly_status, assembly_scanned_at, dispatch_scanned_at	1 Order to N Frames. Used for Assembly and Dispatch QR scanning.
glass_tracker	glass_id, order_id (FK), glass_spec, vendor_name, po_date, expected_delivery_date, actual_received_date, status	Tracks glass delivery schedule independent of profile cutting.
dispatch_logs	dispatch_id, order_id (FK), vehicle_no, driver_phone, whatsapp_alert_sent (BOOLEAN), payment_cleared (BOOLEAN), gate_pass_issued (BOOLEAN)	Manages dispatch loading logs and WhatsApp webhook triggers.

4.2 System Integration Points

- **QR Scanning Interface:** Compatible with 2D Bluetooth Handheld Scanners & Mobile Camera Scanner App.
- **WhatsApp Cloud API:** Webhook service listening to dispatch QR scan events to send automated payment requests.
- **Management Notification Gateway:** High-priority alert push to Uday Sir for rate-pending quote reviews.

5. DEVELOPMENT ROADMAP & SIGN-OFF

PHASE	MODULE	DELIVERABLE & ACCEPTANCE CRITERIA
Phase 1	Order ID & Documents	Auto-generate Canonical Order IDs (e.g., R253 PN); multi-file document upload module operational.
Phase 2	Quote & Rate Approval	Unpriced quote priority queuing active; automated approval alerts sent to Uday Sir.
Phase 3	Shop-Floor QR Assembly	Print unique frame QR stickers; Assembly terminal scanning updates frame status in real time.
Phase 4	Glass Tracking	Glass order dates, expected vendor deliveries, and fitting dates tracked with delay warnings.
Phase 5	Dispatch & WhatsApp API	Dispatch QR scanning triggers WhatsApp payment reminder; Gate Pass hard lock enforced for unpaid balances.

Prepared By (Operations Lead):

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Date: September 21, 2026

Accepted By (IT Technical Lead):

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